

# Linear vs Nonlinear Contrastive Encoders

## Main question

**Does adding a controlled nonlinear correction recover real shared signal, or mostly increase noise fitting?**

## One-line setup

We compare linear and residual nonlinear encoders on noise-only, linear-signal, and cubic-signal synthetic data.

# Synthetic data conditions

Each condition tests a different kind of X/Y relationship while keeping the training protocol fixed.

## Noise only

X and Y are independent Gaussian noise.  
Any training alignment here is memorization.

## Linear signal

X has latent  $Z_x$ , Y has latent  $Z_y$ , with  $Z_y = Z_x$ .

## Cubic signal

X has latent  $Z_x$ , Y has latent  $Z_y$ , with  $Z_y = \text{standardized}(Z_x^3)$ .

## Data scale

Input dimensions:  $X=128$ ,  $Y=128$   
Train pairs: 160  
Held-out pairs: 800

## Signal/noise setting

Latent dimension: 1  
Signal strength: 2.0  
Noise std: 1.0

# Linear backbone plus controlled nonlinear correction

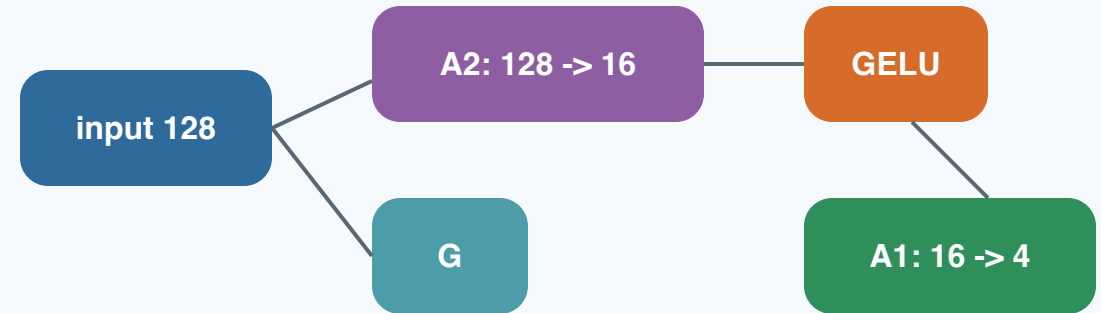
Both views use matched encoder families; residual models use normalized branches so alpha has a meaningful scale.

## Pure linear encoder



$$g(u) = \text{Normalize}(G u)$$

## Residual nonlinear encoder



$$g(u) = \text{Normalize}(G u) + \alpha * \text{Normalize}(A1 \text{ GELU}(A2 u + b))$$

### Current MLP size

Each nonlinear encoder uses one hidden layer: 128 -> 16 -> 4. Embedding dimension is fixed at 4.

# Paper contrastive loss with optional penalties

All model families use the same objective; differences come from architecture and regularization.

## Similarity scores

For a batch of paired embeddings:

$$s_{i,j} = \mathbf{z}_{x,i}^T \mathbf{z}_{y,j}$$

$$\text{maximize } L = \sum_i s_{ii} - \sum_i \log \sum_j \exp(s_{i,j}) - \sum_j \log \sum_i \exp(s_{i,j})$$

training minimizes  $-L / n$  plus any active regularization terms

## L1 penalties

Linear:  $\|\mathbf{G}_x\|_1 + \|\mathbf{G}_y\|_1$

Nonlinear:  $\|\mathbf{A}\|_1$  terms when enabled.

## L2 penalties

Linear:  $\|\mathbf{G}_x\|_2^2 + \|\mathbf{G}_y\|_2^2$

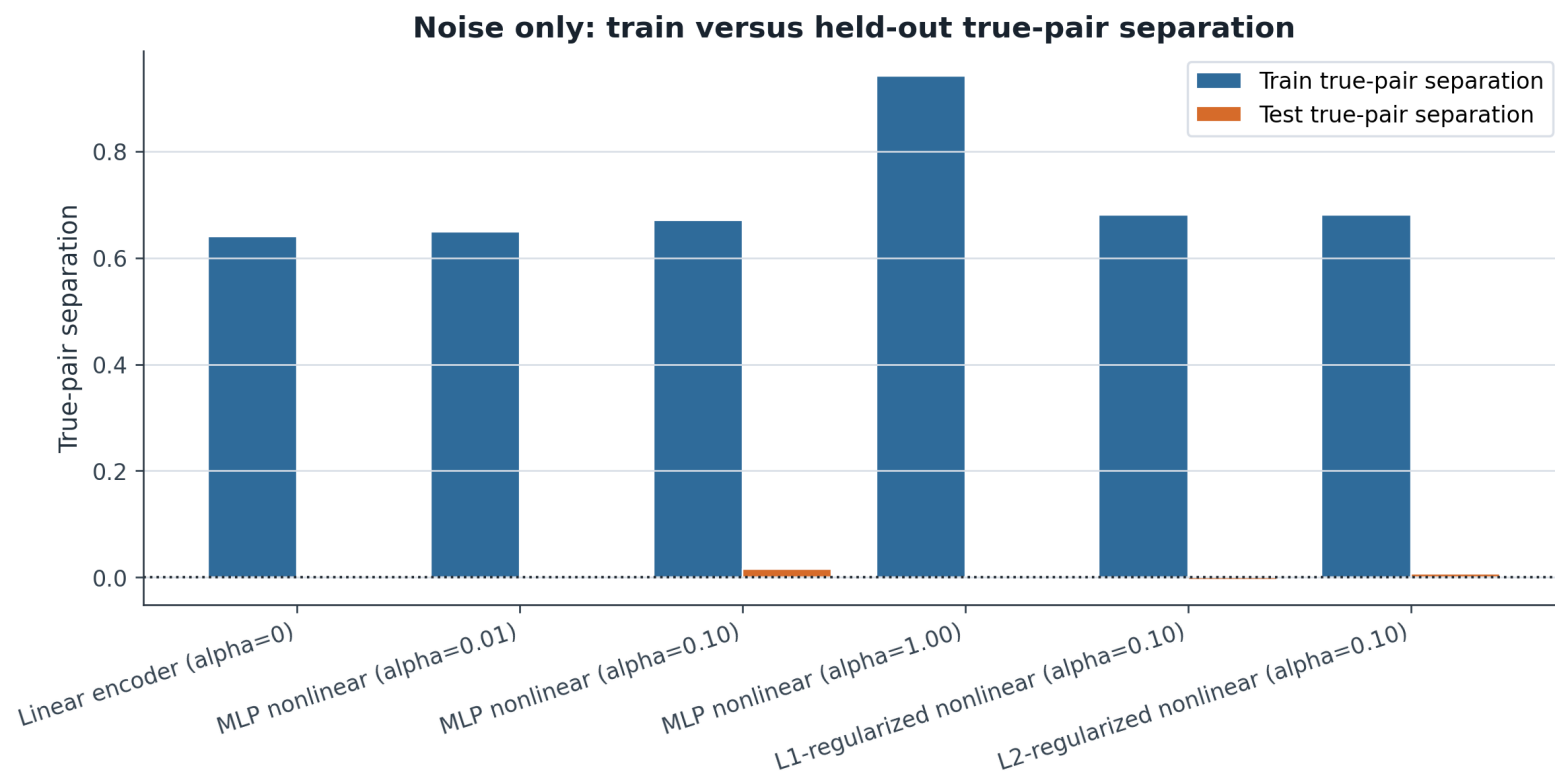
Nonlinear:  $\|\mathbf{A}_1\|_2^2 + \|\mathbf{A}_2\|_2^2$ .

## Output penalty

Available to keep the actual nonlinear correction small unless needed.

# Noise-only: train vs held-out separation

High train separation with low held-out separation indicates memorization rather than signal.



## Interpretation

In noise-only data,  $X$  and  $Y$  have no real relationship. All models can still separate paired training samples, so the training alignment reflects memorized noise pairings. The  $\alpha=1$  nonlinear model shows the strongest training separation, while held-out separation stays near zero for all models.

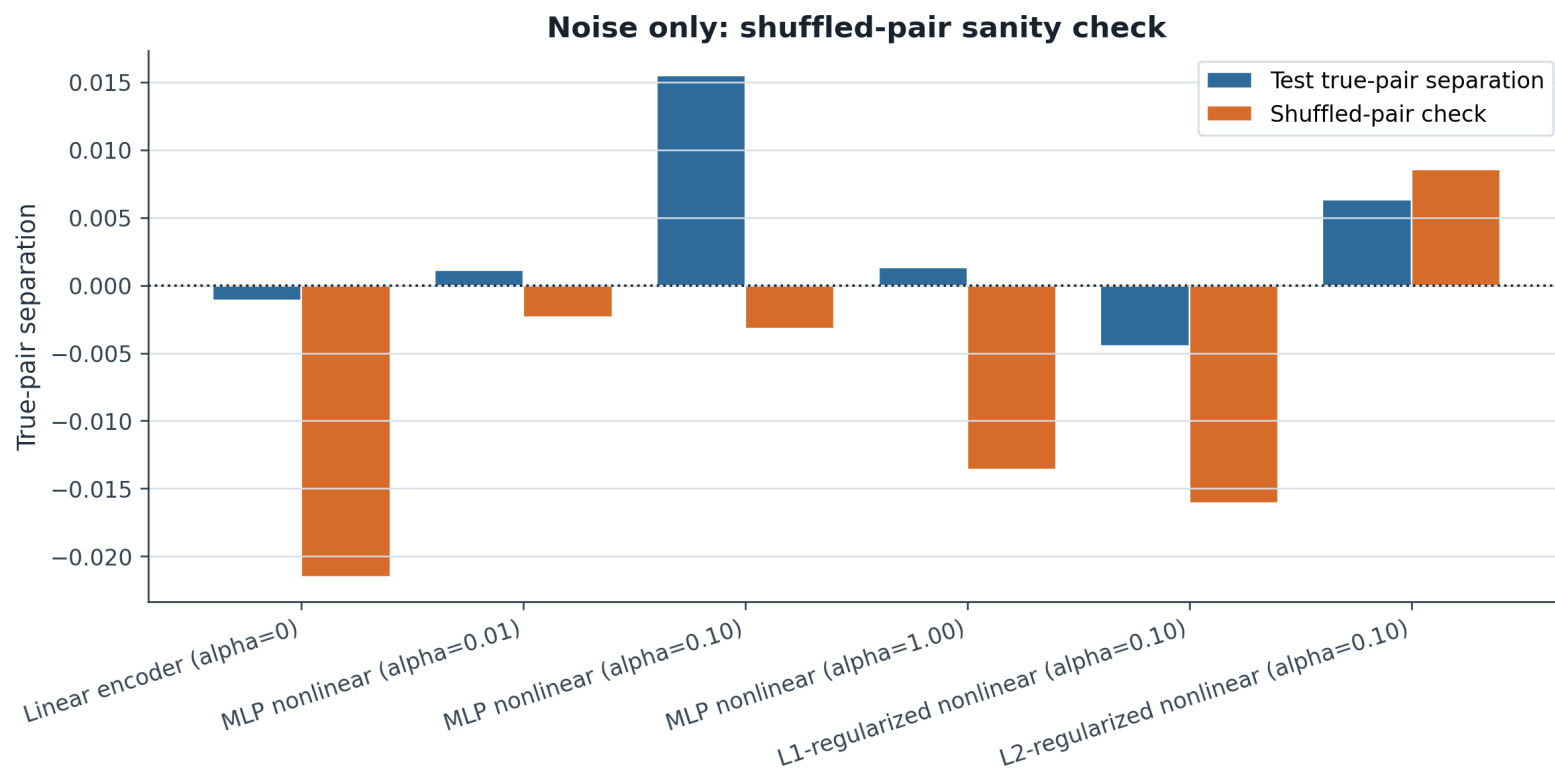
## Key takeaway

**High training alignment in noise does not mean real learning; held-out performance reveals memorization.**

## RESULT

# Noise-only: shuffled-pair sanity check

Breaking the pairing should remove alignment; values near zero support the sanity check.



## Interpretation

This is a sanity check for noise-only data. Since  $X$  and  $Y$  are independent, true held-out pairs should not be more meaningful than shuffled pairs. Both original held-out separation and shuffled-pair separation stay close to zero.

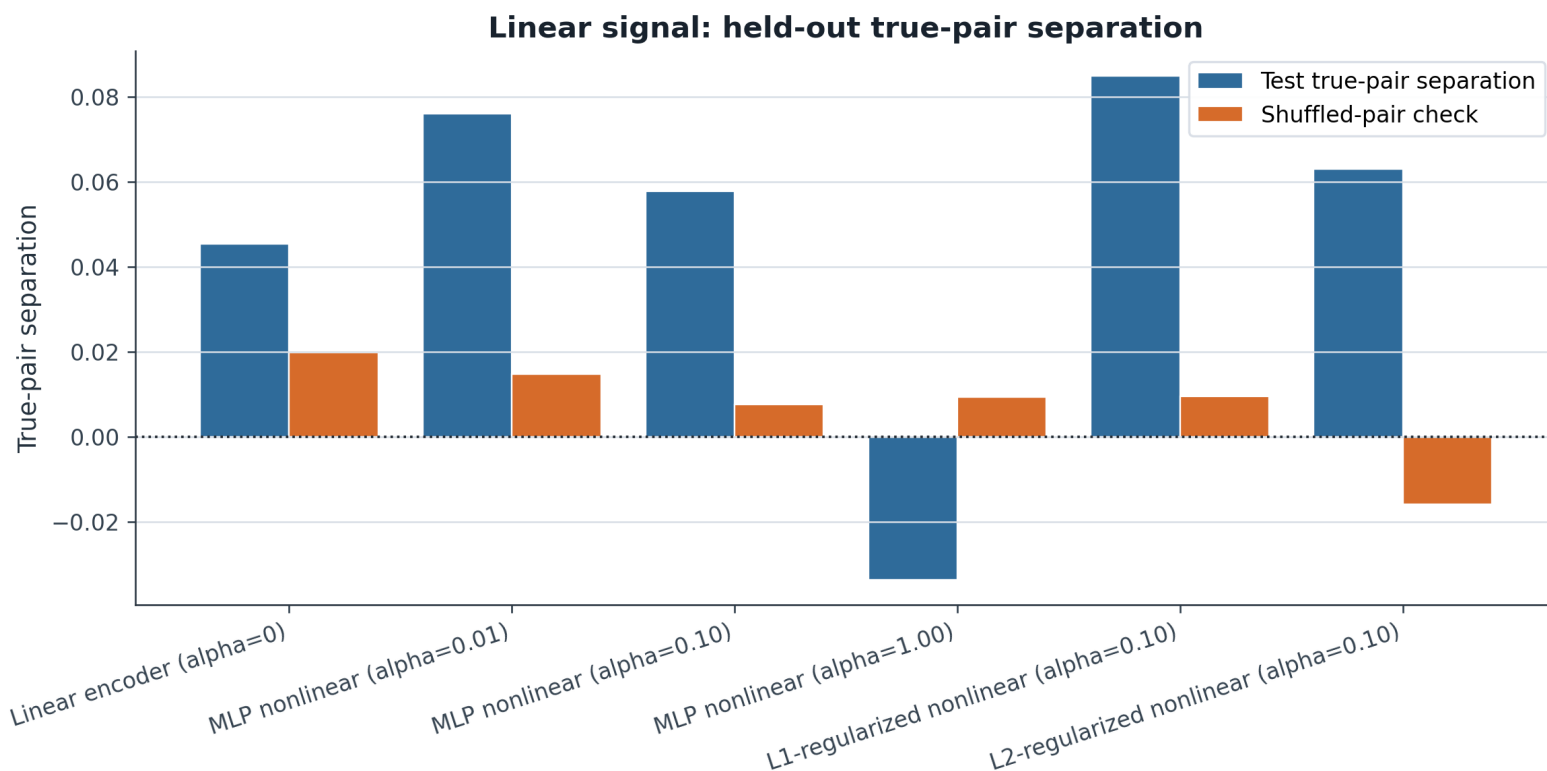
## Key takeaway

**True held-out pairs behave like shuffled random pairs, confirming that no real signal is learned.**

## RESULT

# Linear signal: held-out separation

The linear baseline should already be competitive when  $Z_y = Z_x$ .



## Interpretation

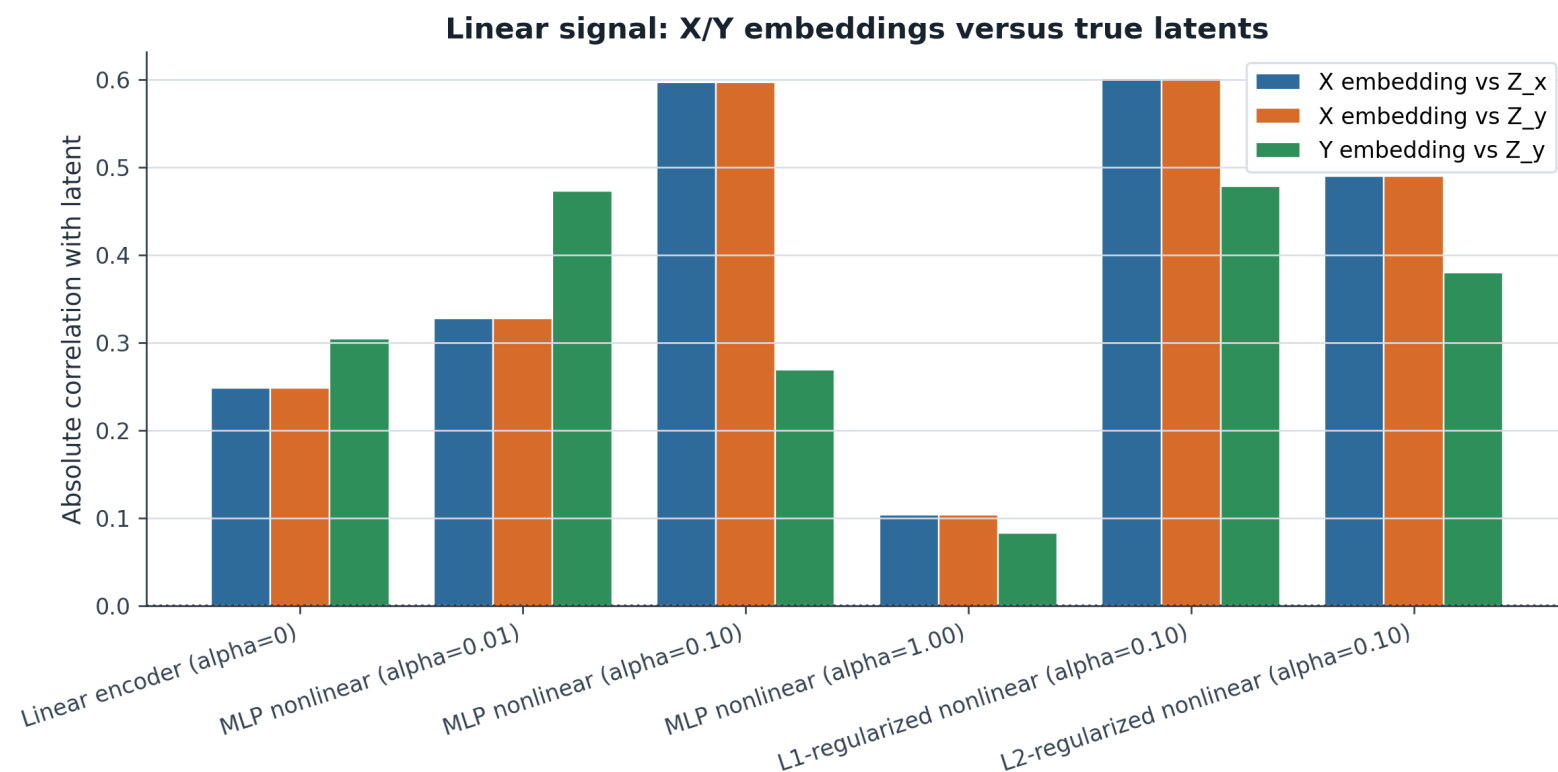
With linear signal, held-out true-pair separation is positive for most models, so they capture shared structure that generalizes. The linear baseline is already competitive, shuffled values are smaller, and alpha=1 performs poorly.

## Key takeaway

**When the true relationship is linear, the linear baseline is strong and a dominant nonlinear branch does not necessarily help.**

# Linear signal: latent recovery

Correlation with the true latent shows whether embeddings preserve the synthetic signal.



## Interpretation

This plot measures how much true latent signal remains in the embeddings. Moderate nonlinear contribution and the L2-regularized nonlinear model improve X-side recovery, while  $\alpha=1$  hurts recovery. Y-side recovery is more uneven.

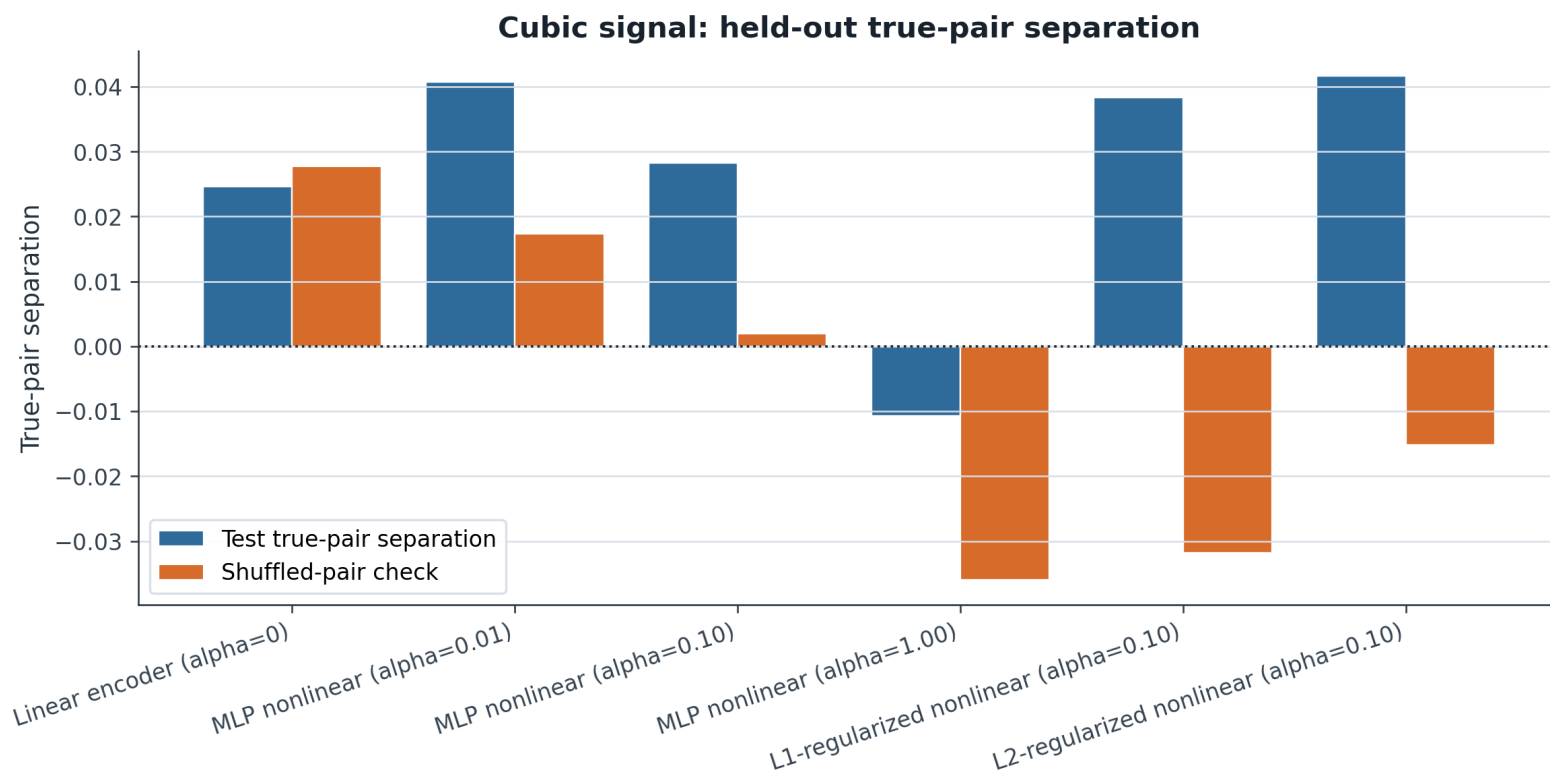
## Key takeaway

**Moderate nonlinear correction can improve latent recovery, but too much nonlinear influence can hurt even for linear signal.**



# Cubic signal: held-out separation

The cubic relationship asks whether the nonlinear correction helps beyond the linear backbone.



## Interpretation

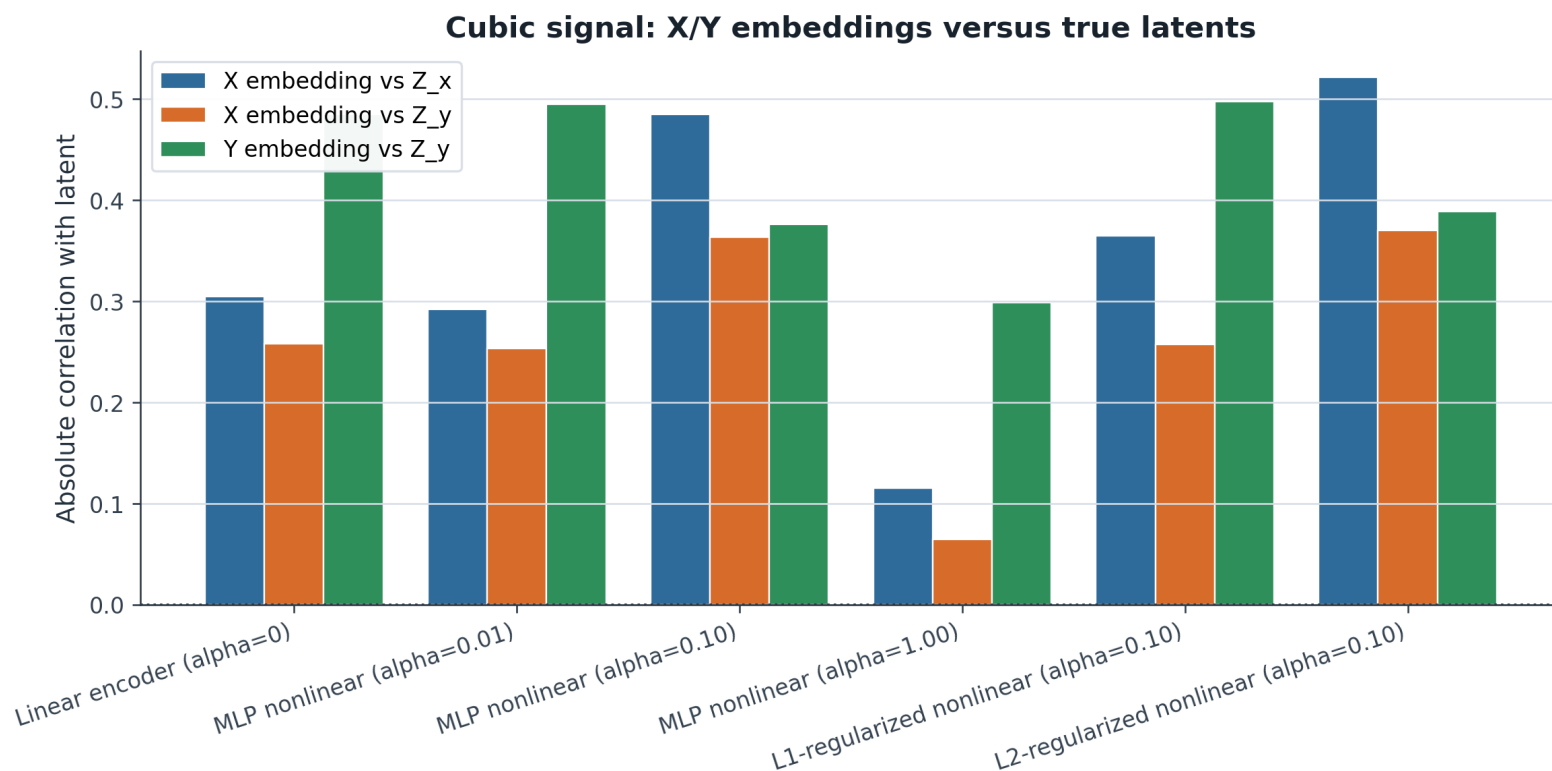
The cubic setting tests whether nonlinear correction helps beyond the linear backbone. Some controlled nonlinear or regularized models improve held-out separation over the shuffled baseline, suggesting partial nonlinear structure recovery. The  $\alpha=1$  model generalizes poorly.

## Key takeaway

**Controlled nonlinear corrections can help nonlinear signal, but an overly large nonlinear branch can destabilize learning.**

# Cubic signal: latent recovery

The recovery plot asks whether embeddings capture the true nonlinear latent relationship.



## Interpretation

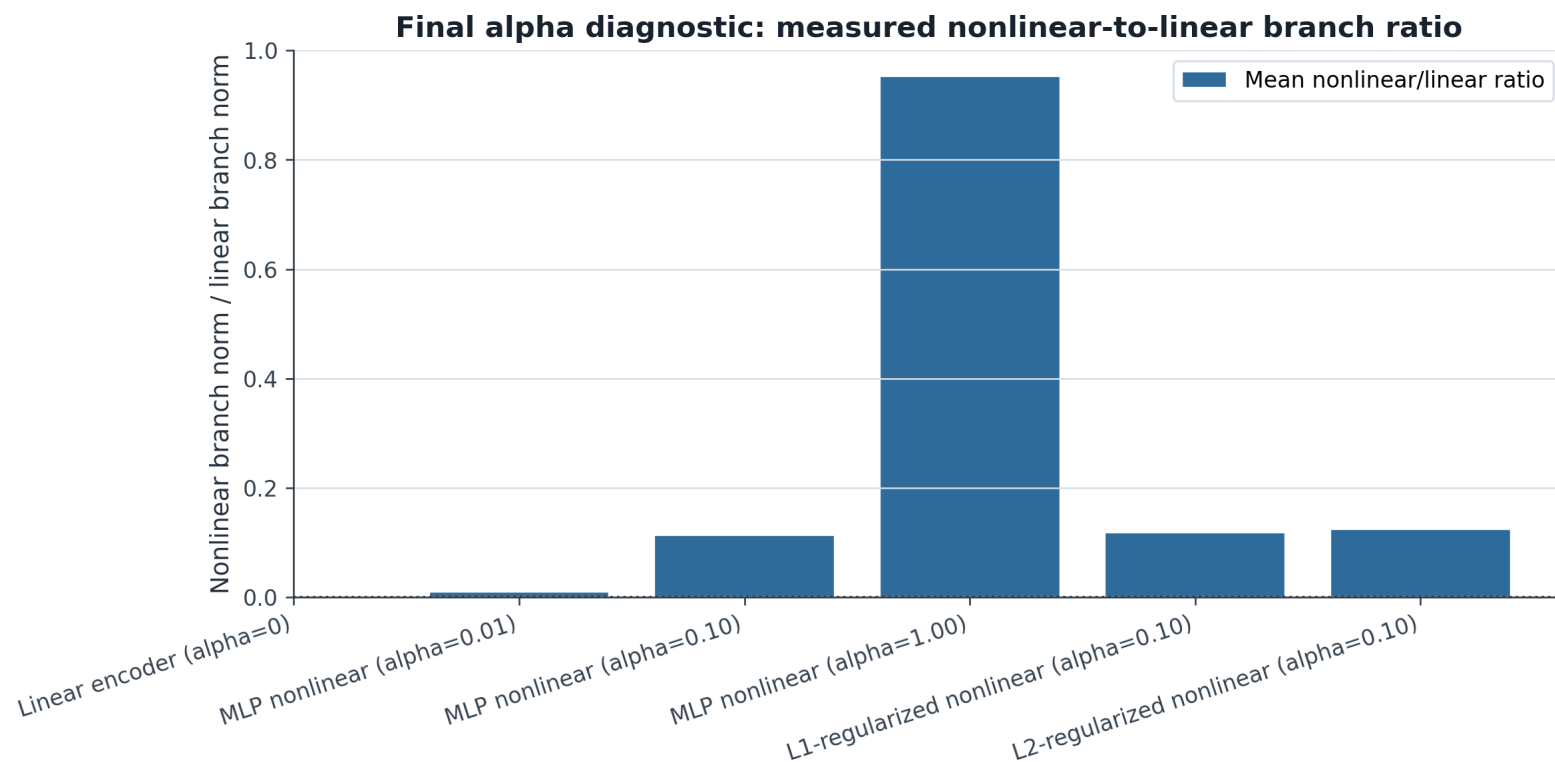
Latent recovery is less consistent than in the linear setting, showing the cubic relationship is harder to capture. Controlled nonlinear or regularized models preserve more latent information for some views, while  $\alpha=1$  damages the representation.

## Key takeaway

**Nonlinear structure is harder to recover; controlled nonlinear correction helps more than a dominant nonlinear branch.**

# Branch ratio diagnostic

The measured nonlinear/linear branch ratio checks whether alpha controls the correction size.



## Interpretation

This diagnostic checks whether alpha controls the nonlinear correction size. Because both branches are normalized before summing, the nonlinear-to-linear ratio should track alpha: alpha=0.01 is tiny, alpha=0.10 is controlled, and alpha=1 is nearly as large as the linear branch.

## Key takeaway

**Branch normalization makes alpha interpretable: small alpha gives a controlled correction, while alpha=1 lets the nonlinear branch dominate.**

# Findings and next steps

The cleaned experiment separates signal recovery from noise fitting.

## Findings to report

Noise-only: high train alignment but held-out near zero means memorization.

Linear signal: the linear baseline is strong; moderate nonlinear correction can improve recovery.

Cubic signal: controlled or regularized nonlinear models help more than  $\alpha=1$ .

## Main caution

The train/test alignment gap is still large in the linear and cubic settings. Branch normalization makes  $\alpha$  interpretable, but it does not by itself solve overfitting or loss-design issues.

## Next steps

Understand and reduce the train/test alignment gap for linear and cubic signals.

Try different contrastive loss functions.

Tune the early stopping mechanism and stopping criteria.

Try other signal relationships beyond linear and cubic.

Repeat the strongest settings across more random seeds.